

Advanced-statistics-project(P8-G4)

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1 Allometric scaling of metabolic rate: is Kleiber’s law universal?

Advanced Statistics for Physics Analysis — Project P08 (Bayesian inference)

Group P8-G4

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Basal metabolic rate B is usually assumed to follow a power law in body mass M ,

$$B \propto M^b.$$

Kleiber’s law fixes the exponent at $b = 3/4$. Whether a single $3/4$ exponent really holds for *all* animals — or even for mammals and birds separately — has been argued about for almost a century; the older “surface law” predicts $b = 2/3$ instead. The point of this project is to put a number, *with uncertainty*, on that exponent and to test the $3/4$ hypothesis directly.

We work in log–log space, with $x_i = \ln M_i$ and $y_i = \ln B_i$, so a power law becomes a straight line

$$y_i = \alpha + bx_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$

and the slope of that line *is* the allometric exponent. We fit three Bayesian regression models with a **hand-coded Gibbs sampler** (and cross-check the result against **JAGS**), check convergence with **coda**, compare the models with the DIC, and finish with posterior-predictive checks and a prediction for a 1 kg animal.

The three questions we answer: 1. What is the posterior distribution of the exponent β (free-slope model)? 2. Is Kleiber’s $3/4$ compatible with the data — i.e. what is $P(\beta > 3/4 \mid \text{data})$ and is $3/4$ inside the 95% credible interval? 3. Is one exponent enough, or do mammals and birds scale differently?

1.1 1. Data and preparation

We use the **AnimalTraits** database (Herberstein *et al.*, *Scientific Data* **9**, 265, 2022), a curated open database of terrestrial-animal body mass, metabolic rate and brain size in standardised units (Zenodo record 6468938).

Starting from the standardised **observations.csv**, we kept only rows that have **both** a body mass and a whole-body metabolic rate, and — as the brief asks — only those whose **metabolic rate - method** is **basal metabolic rate**. We then converted metabolic rate to watts and body mass to kilograms using the database’s own conventions (oxygen consumption converted with 20 J ml^{-1} ,

basal rates of endotherms are *not* temperature-corrected). One practical consequence of the “basal” filter is that it essentially selects endotherms: the surviving records are all **mammals and birds** (for ectotherms one speaks of *standard*, not *basal*, metabolic rate). The taxonomic group used in Model 3 is therefore the class — *Mammalia* vs *Aves*.

The cleaned analysis table is stored next to this notebook as `animaltraits_bmr.csv`.

```
[20]: library(rjags)
library(coda)
set.seed(200)

# Data selection (reproduces the brief's procedure from the raw AnimalTraits
# file)
# Keep rows whose metabolic-rate method is "basal metabolic rate" and that have
# BOTH a
# body mass and a metabolic rate; use the standardized columns (metabolic rate
# in W,
# body mass in kg). The basal filter leaves only endotherms (birds + mammals).
if (file.exists("observations.csv")) {
  raw <- read.csv("observations.csv", check.names = FALSE, stringsAsFactors =
  FALSE)
  sel <- trimws(raw[["metabolic rate - method"]]) == "basal metabolic rate" &
    !is.na(raw[["body mass"]]) & !is.na(raw[["metabolic rate"]])
  dat <- data.frame(species      = raw$species[sel],
                    taxon_group = raw$class[sel],
                    body_mass_kg = raw[["body mass"]][sel],      #
    standardized, kg      metabolic_rate_W = raw[["metabolic rate"]][sel], #
    standardized, W      stringsAsFactors = FALSE)
  dat <- dat[dat$taxon_group %in% c("Aves", "Mammalia"), ]
  write.csv(dat, "animaltraits_bmr.csv", row.names = FALSE)      # cache
  cleaned table
} else {
  dat <- read.csv("animaltraits_bmr.csv", stringsAsFactors = FALSE) # offline
  fallback
}

dat$x <- log(dat$body_mass_kg)      # ln body mass (kg)
dat$y <- log(dat$metabolic_rate_W)  # ln basal metabolic rate (W)
dat$grp <- as.integer(factor(dat$taxon_group))  # 1 = Aves, 2 = Mammalia

cat("N =", nrow(dat), "\n")
print(table(dat$taxon_group))
cat(sprintf("mass range: %.4g - %.4g kg\n", min(dat$body_mass_kg),
  max(dat$body_mass_kg)))
```

N = 279

```

      Aves Mammalia
      133      146
mass range: 0.0027 - 48 kg

```

```

[21]: head(dat[,  
           ↪c("species", "taxon_group", "body_mass_kg", "metabolic_rate_W", "x", "y")], 5)

```

```

A data.frame: 5 × 6

```

	species <chr>	taxon_group <chr>	body_mass_kg <dbl>	metabolic_rate_W <dbl>	x <dbl>	y <dbl>
1	Psaltiriparus minimus	Aves	0.0055	0.121000	-5.2030072	-2.1119647
2	Amazona viridigenalis	Aves	0.3419	2.013333	-1.0732370	0.6997917
3	Anas crecca	Aves	0.2500	1.666667	-1.3862944	0.5108256
4	Anas querquedula	Aves	0.2890	2.233333	-1.2413286	0.8034952
5	Aythya nyroca	Aves	0.4400	3.272222	-0.8209806	1.1854693

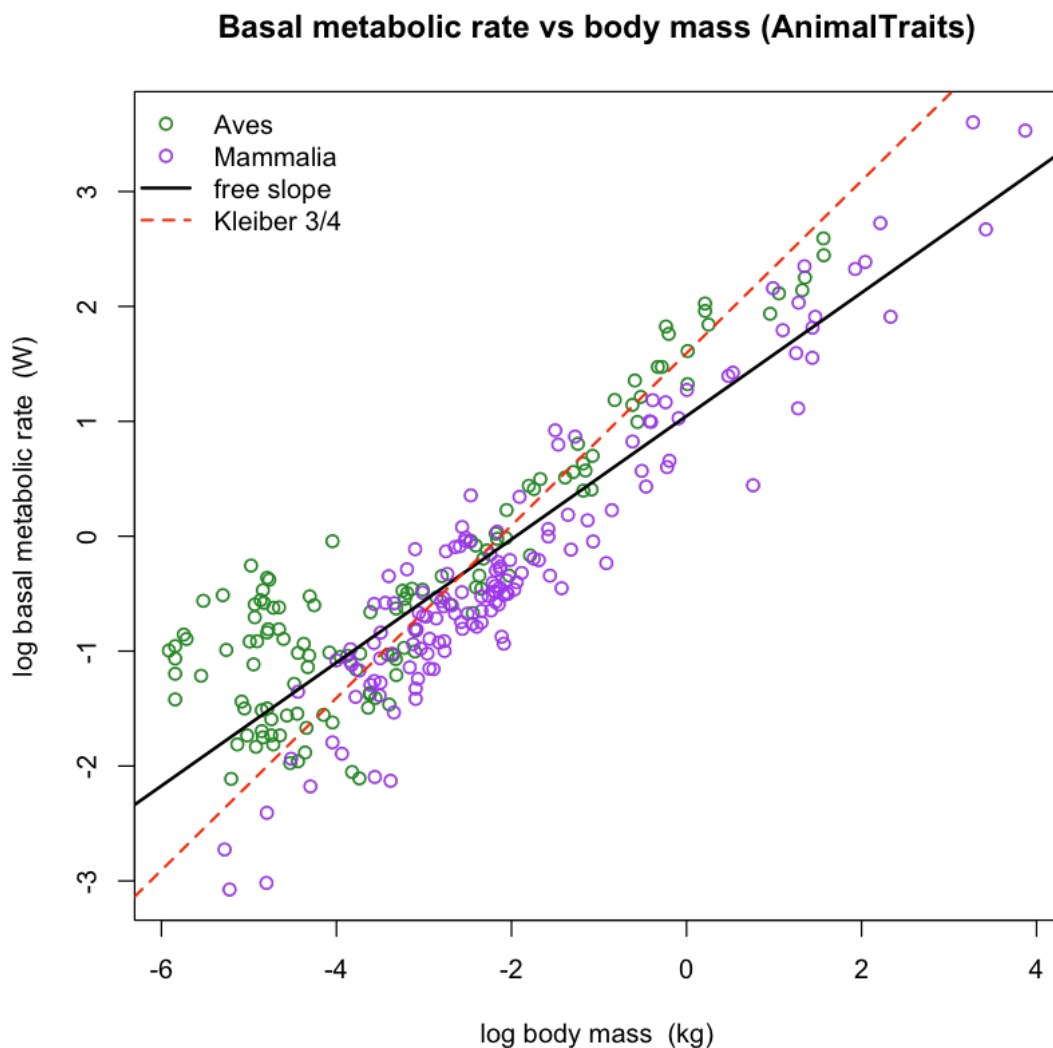
1.2 2. Exploratory plot

A first look on log-log axes, colouring the two classes. The dashed red line is Kleiber's prediction (slope 3/4), its intercept is taken from the fixed-exponent fit below so that only the *slope* is being compared.

```

[22]: cols <- c(Aves = "forestgreen", Mammalia = "purple")
plot(dat$x, dat$y, pch = 1, lwd = 1.3, col = cols[dat$taxon_group],
      xlab = "log body mass (kg)", ylab = "log basal metabolic rate (W)",
      main = "Basal metabolic rate vs body mass (AnimalTraits)")
abline(lm(y ~ x, data = dat), lwd = 2) # free slope
abline(a = 1.593, b = 0.75, col = "red", lty = 2, lwd = 1.6) # Kleiber 3/4
legend("topleft", bty = "n",
      legend = c("Aves", "Mammalia", "free slope", "Kleiber 3/4"),
      col = c("forestgreen", "purple", "black", "red"),
      pch = c(1, 1, NA, NA), lty = c(NA, NA, 1, 2), lwd = c(1.3, 1.3, 2, 1.6))

```



Two things already stand out. The cloud is clearly linear over more than four orders of magnitude in mass, so a power law is a reasonable description. But the overall slope is visibly *shallower* than the red $3/4$ line, and at a given mass the birds (green) tend to sit *above* the mammals (purple) — a hint that the two classes differ mainly in *elevation*. We now make this quantitative.

1.3 3. Models, likelihood and priors

All three models share the Gaussian likelihood

$$y_i \sim \mathcal{N}(\mu_i, \sigma^2),$$

and differ only in the mean function μ_i :

- **Model 1 — Kleiber, fixed exponent:** $\mu_i = \alpha + \frac{3}{4}x_i$.
- **Model 2 — free exponent:** $\mu_i = \alpha + \beta x_i$.

- **Model 3 — group-dependent scaling:** $\mu_i = \alpha_{g[i]} + \beta_{g[i]} x_i$, where $g[i] \in \{\text{Aves, Mammalia}\}$.

Priors. We use deliberately weak priors so the data dominate:

$$\alpha, \beta \sim \mathcal{N}(0, 100^2), \quad \tau = 1/\sigma^2 \sim \text{Gamma}(0.001, 0.001).$$

(JAGS parameterises the normal by its *precision*, so $\mathcal{N}(0, 100^2)$ is written `dnorm(0, 1.0E-4)`.) The Gamma prior on the precision is the usual vague choice that keeps $\sigma > 0$.

Our primary engine is a hand-coded Gibbs sampler (4 chains of 20,000 iterations, 4,000 discarded as burn-in); the JAGS cross-check in §7 uses 4 chains with 2000 adaptation + 2000 burn-in + 5000 sampling iterations.

1.3.1 3.1 A hand-coded Gibbs sampler

Rather than rely on a black box, we implement the sampler ourselves. All three models are Gaussian linear models $y = Xb + \varepsilon$ with conjugate priors $b \sim N(0, \lambda_0^{-1}I)$ and $\tau = 1/\sigma^2 \sim \text{Gamma}(a_0, b_0)$, so every parameter has a closed-form full conditional and a **Gibbs sampler** is exact — no tuning, no rejection step:

$$b \mid \tau, y \sim N(V \tau X^\top y, V), \quad V = (\lambda_0 I + \tau X^\top X)^{-1}, \quad \tau \mid b, y \sim \text{Gamma}(a_0 + \frac{n}{2}, b_0 + \frac{1}{2} \|y - Xb\|^2).$$

Because the routine takes an arbitrary design matrix X , the *same* code fits all three models: Model 1 (intercept only, with the slope folded in via the offset $y - \frac{3}{4}x$), Model 2 ($X = [1, x]$), and Model 3 (group-coded columns). We use the vague priors $\lambda_0 = 10^{-4}$ (i.e. $\alpha, \beta \sim N(0, 100^2)$) and $a_0 = b_0 = 10^{-3}$ — identical to the JAGS models cross-checked in §7.

```
[23]: # Hand-coded Gibbs sampler for a Gaussian linear model
# y ~ N(X b, sigma^2 I); b ~ N(0, lambda0^{-1} I); tau = 1/sigma^2 ~
# Gamma(a0, b0)
# Full conditionals (conjugate):
# b | tau, y ~ N( V (tau X'y), V ), V = (lambda0 I + tau X'X)^{-1}
# tau | b, y ~ Gamma( a0 + n/2, b0 + (1/2) ||y - X b||^2 )
gibbs_lm <- function(X, y, n.iter = 20000, burn = 4000,
                     lambda0 = 1e-4, a0 = 1e-3, b0 = 1e-3, seed = 1) {
  set.seed(seed)
  X <- as.matrix(X); n <- nrow(X); k <- ncol(X)
  XtX <- crossprod(X); Xty <- crossprod(X, y)
  b <- rep(0, k); tau <- 1
  keep <- matrix(NA, n.iter, k + 1)
  for (t in 1:n.iter) {
    V <- solve(lambda0 * diag(k) + tau * XtX) # coefficient covariance
    m <- V %*% (tau * Xty) # coefficient mean
    b <- as.vector(m + t(chol(V)) %*% rnorm(k)) # draw b ~ N(m, V)
    rss <- sum((y - X %*% b)^2)
    tau <- rgamma(1, a0 + n/2, b0 + rss/2) # draw tau
    keep[t, ] <- c(b, 1 / sqrt(tau))
  }
}
```

```

}
colnames(keep) <- c(paste0("b", 1:k), "sigma")
keep[(burn + 1):n.iter, , drop = FALSE] # discard burn-in
}
cat("gibbs_lm() ready")

```

gibbs_lm() ready

1.3.2 3.2 Fitting the three models

Each model runs for 20,000 iterations with the first 4,000 discarded as burn-in. The columns `b1`, `b2`, ... are the regression coefficients (for Model 3: Aves intercept, Mammalia intercept, Aves slope, Mammalia slope) and `sigma` is the residual standard deviation.

```

[24]: # Design matrices for the three models
X1 <- matrix(1, nrow(dat), 1) # Model 1: intercept only
# (slope fixed at 3/4)
y1 <- dat$y - 0.75 * dat$x # fit the offset
# y - 3/4 x
X2 <- cbind(1, dat$x) # Model 2: alpha + beta x
gA <- as.numeric(dat$grp == 1); gM <- as.numeric(dat$grp == 2)
X3 <- cbind(gA, gM, dat$x * gA, dat$x * gM) # Model 3: per-group
# intercept & slope

h1 <- gibbs_lm(X1, y1, seed = 11) # cols: b1 = alpha, sigma
h2 <- gibbs_lm(X2, dat$y, seed = 212) # b1 = alpha, b2 = beta,
# sigma
h3 <- gibbs_lm(X3, dat$y, seed = 67) # b1=aAves, b2=aMam,
# b3=bAves, b4=bMam, sigma

ci <- function(z) quantile(z, c(.025, .975))
cat(sprintf("Model 1 (fixed 3/4): alpha = %.3f [%.3f, %.3f] sigma = %.3f\n",
  mean(h1[, "b1"]), ci(h1[, "b1"])[1], ci(h1[, "b1"])[2], mean(h1[, "sigma"])))
cat(sprintf("Model 2 (free): alpha = %.3f [%.3f, %.3f] beta = %.3f [%.
  3f, %.3f] sigma = %.3f\n",
  mean(h2[, "b1"]), ci(h2[, "b1"])[1], ci(h2[, "b1"])[2],
  mean(h2[, "b2"]), ci(h2[, "b2"])[1], ci(h2[, "b2"])[2], mean(h2[, "sigma"])))
cat(sprintf("Model 3 Aves: alpha = %.3f [%.3f, %.3f] beta = %.3f [%.
  3f, %.3f]\n",
  mean(h3[, "b1"]), ci(h3[, "b1"])[1], ci(h3[, "b1"])[2], mean(h3[, "b3"]),
  ci(h3[, "b3"])[1], ci(h3[, "b3"])[2]))
cat(sprintf("Model 3 Mammalia: alpha = %.3f [%.3f, %.3f] beta = %.3f [%.
  3f, %.3f] sigma = %.3f\n",
  mean(h3[, "b2"]), ci(h3[, "b2"])[1], ci(h3[, "b2"])[2],
  mean(h3[, "b4"]), ci(h3[, "b4"])[1], ci(h3[, "b4"])[2], mean(h3[, "sigma"])))

```

```

Model 1 (fixed 3/4): alpha = 1.593 [1.515, 1.671] sigma = 0.672
Model 2 (free): alpha = 1.045 [0.941, 1.148] beta = 0.536 [0.504, 0.568]

```

```
sigma = 0.528
Model 3 Aves:      alpha = 1.255 [1.096, 1.410]    beta = 0.527 [0.486, 0.569]
Model 3 Mammalia:  alpha = 1.014 [0.901, 1.125]    beta = 0.630 [0.588, 0.672]
sigma = 0.463
```

1.3.3 3.3 Convergence diagnostics

To confirm the hand-coded sampler has converged we run four independent chains from different seeds and inspect the Gelman–Rubin $\hat{R}$, the effective sample sizes, and the trace/density of β . (coda is used here only to *diagnose*; the draws themselves come entirely from our own sampler. Because the coefficient vector is updated as one joint block, the chain mixes almost perfectly.)

```
[25]: #Convergence of the hand-coded sampler: 4 independent chains for Model 2
chains <- mcmc.list(lapply(c(11, 22, 33, 44), function(s)
  mcmc(gibbs_lm(X2, dat$y, seed = s)[, c("b1", "b2", "sigma")])))
varnames(chains) <- c("alpha", "beta", "sigma")
print(gelman.diag(chains))                # Gelman-Rubin R-hat
cat("\nEffective sample size:\n"); print(round(effectiveSize(chains)))
plot(chains[, "beta"])                    # trace + density of beta
```

Potential scale reduction factors:

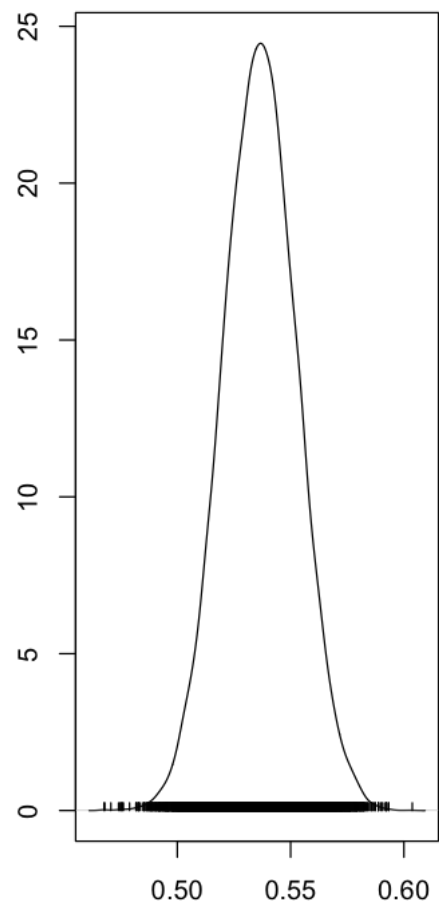
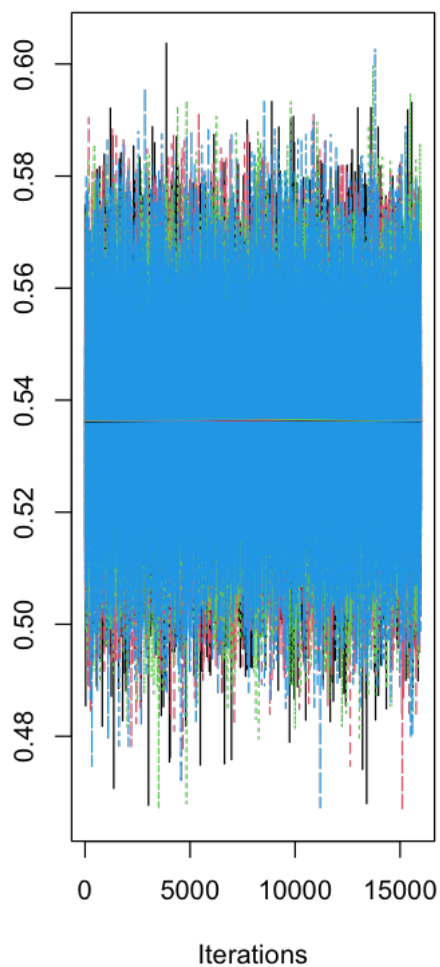
	Point est.	Upper C.I.
alpha	1	1
beta	1	1
sigma	1	1

Multivariate psrf

1

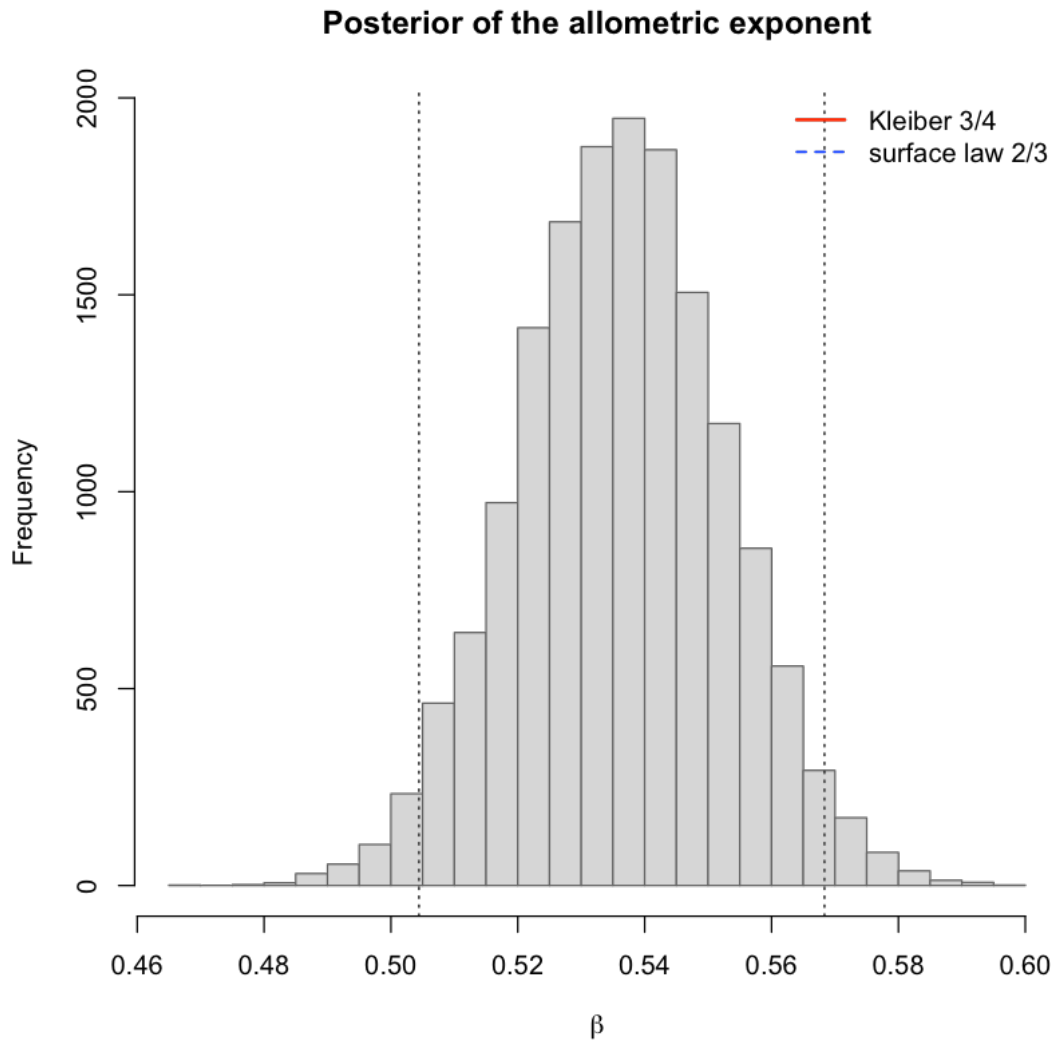
Effective sample size:

alpha	beta	sigma
63869	63651	62286



1.3.4 3.4 Posterior of the exponent and the Kleiber test

```
[26]: B <- h2[, "b2"] # posterior draws of the exponent (hand-coded Gibbs)
hist(B, breaks = 45, col = "grey85", border = "grey40",
     xlab = expression(beta), main = "Posterior of the allometric exponent")
abline(v = 0.75, col = "red", lwd = 2) # Kleiber 3/4
abline(v = 2/3, col = "blue", lwd = 1.4, lty = 2) # surface law 2/3
abline(v = quantile(B, c(.025, .975)), lty = 3)
legend("topright", bty = "n", col = c("red", "blue"), lwd = c(2, 1.4), lty = c(1, 2),
      legend = c("Kleiber 3/4", "surface law 2/3"))
```

```
[27]: cat(sprintf("Posterior mean of beta : %.3f\n", mean(B)))
cat(sprintf("95% credible interval : [%.3f, %.3f]\n", quantile(B,.025),
↳ quantile(B,.975)))
cat(sprintf("P(beta > 3/4 | data) : %.3f\n", mean(B > 0.75)))
cat(sprintf("P(beta > 2/3 | data) : %.3f\n", mean(B > 2/3)))
```

```
Posterior mean of beta : 0.536
95% credible interval : [0.504, 0.568]
P(beta > 3/4 | data) : 0.000
P(beta > 2/3 | data) : 0.000
```

The posterior for β is centred at **0.54**, with a tight 95% credible interval [**0.50, 0.57**]. Kleiber's 3/4 is far outside this interval, so $P(\beta > 3/4 \mid \text{data}) = 0$ to three decimals — and the same holds for the 2/3 surface law ($P(\beta > 2/3) = 0$). On the pooled basal sample the exponent is *significantly*

shallower than either classical value.

1.3.5 3.5 Model 3 — group-dependent scaling

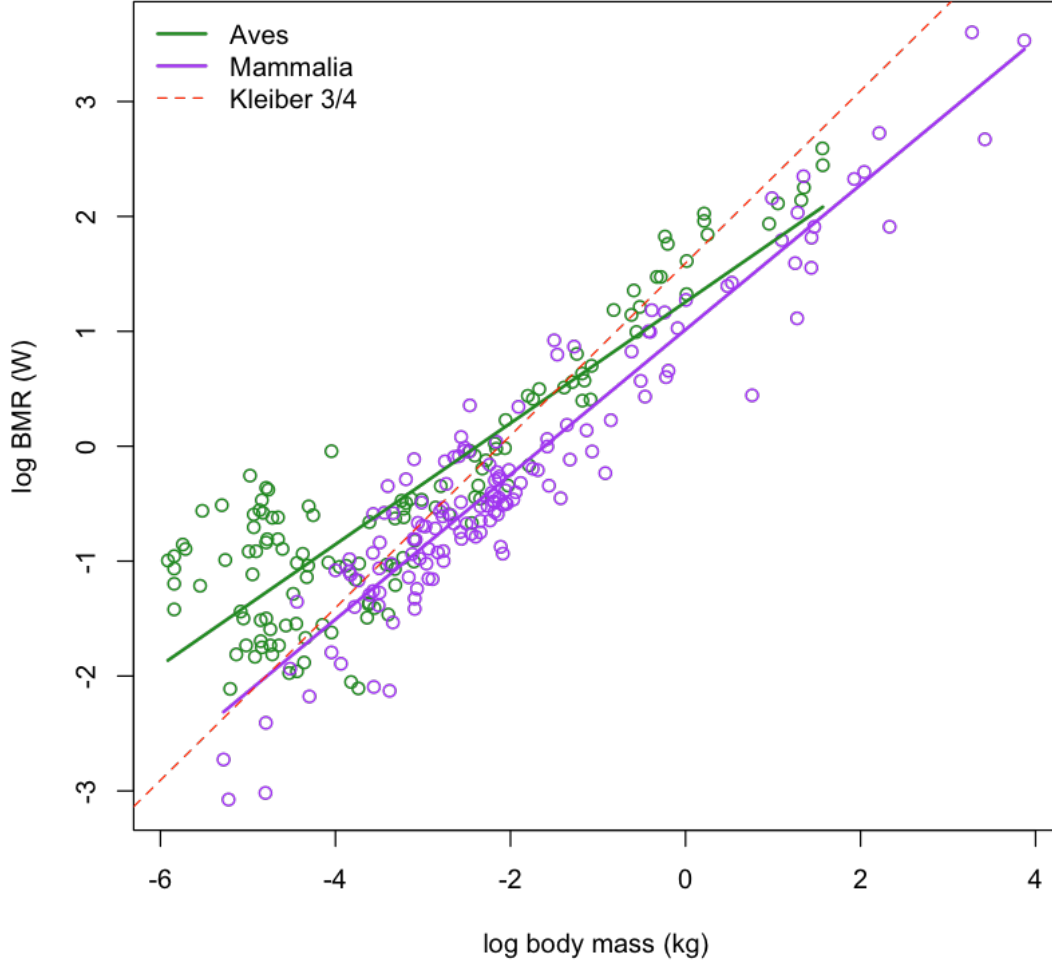
Now each class gets its own intercept and slope (b1,b2 are the Aves/Mammalia intercepts and b3,b4 the Aves/Mammalia slopes in h3). We report each group's slope, the probability it exceeds 3/4, and the within-group regression lines.

```
[28]: grpname <- c("Aves", "Mammalia")
for (g in 1:2) {
  bg <- h3[, paste0("b", g + 2)]          # b3 = Aves slope, b4 = Mammalia slope
  cat(sprintf("%-9s beta = %.3f  95%% CrI [%.3f, %.3f]  P(beta>3/4) = %.3f\n",
    grpname[g], mean(bg), quantile(bg,.025), quantile(bg,.975), mean(bg > 0.75)))
}
cat(sprintf("P(beta_Mammalia > beta_Aves) = %.3f\n", mean(h3[, "b4"] > h3[, "b3"])))
```

```
Aves      beta = 0.527  95% CrI [0.486, 0.569]  P(beta>3/4) = 0.000
Mammalia  beta = 0.630  95% CrI [0.588, 0.672]  P(beta>3/4) = 0.000
P(beta_Mammalia > beta_Aves) = 1.000
```

```
[29]: plot(dat$x, dat$y, pch = 1, lwd = 1.2, col = cols[dat$taxon_group],
  xlab = "log body mass (kg)", ylab = "log BMR (W)", main = "Group-dependent scaling")
for (g in 1:2) {
  s <- dat[dat$grp == g, ]
  ag <- mean(h3[, paste0("b", g)]); bg <- mean(h3[, paste0("b", g + 2)])
  xs <- seq(min(s$x), max(s$x), length = 50)
  lines(xs, ag + bg * xs, col = c("forestgreen", "purple")[g], lwd = 2)
}
abline(a = 1.593, b = 0.75, col = "red", lty = 2)
legend("topleft", bty = "n", legend = c("Aves", "Mammalia", "Kleiber 3/4"),
  col = c("forestgreen", "purple", "red"), lwd = c(2,2,1), lty = c(1,1,2))
```

Group-dependent scaling



On their own the two classes have **different** slopes: mammals scale as $\beta_{\text{Mam}} \approx \mathbf{0.63}$ and birds as $\beta_{\text{Aves}} \approx \mathbf{0.53}$. Mammals are reliably the steeper of the two — $P(\beta_{\text{Mam}} > \beta_{\text{Aves}}) \approx \mathbf{1.00}$ — yet *both* slopes are still clearly below Kleiber’s 3/4. The classes also differ in *elevation*: at the same mass a bird burns more energy than a mammal ($\alpha_{\text{Aves}} = 1.25$ vs $\alpha_{\text{Mam}} = 1.01$). Because the birds (high elevation) populate the small-mass end and the mammals the large-mass end, pooling these two offset grades into a single line in Model 2 tilts the common slope *down* to ≈ 0.54 — shallower even than the mammal slope on its own. This aggregation effect is exactly why the group-structured model fits so much better, as the formal comparison now confirms.

1.4 4. Model comparison (DIC)

The Deviance Information Criterion trades goodness of fit against the effective number of parameters p_D ; lower is better. We compute it **directly from the posterior draws** of each model — Spiegelhalter’s $\text{DIC} = \bar{D} + p_D$ with $p_D = \bar{D} - D(\hat{\theta})$ — so nothing is hidden inside a library routine.

```
[30]: # DIC straight from the posterior draws:  $DIC = \bar{D} + pD$ ,  $pD = \bar{D} -$ 
      ↪  $D(\hat{\theta})$ 
dic_lm <- function(draws, X = NULL, y, fixed = NULL, x = NULL) {
  S <- nrow(draws)
  dev <- function(mu, sg) -2 * sum(dnorm(y, mu, sg, log = TRUE))
  k <- if (is.null(fixed)) ncol(X) else 1
  D <- numeric(S)
  for (s in 1:S) {
    mu <- if (is.null(fixed)) as.vector(X %>% draws[s, 1:k]) else draws[s, 1] +
    ↪ fixed * x
    D[s] <- dev(mu, draws[s, "sigma"])
  }
  muhat <- if (is.null(fixed)) as.vector(X %>% colMeans(draws[, 1:k, drop =
  ↪ FALSE]))
    else mean(draws[, 1]) + fixed * x
  Dhat <- dev(muhat, mean(draws[, "sigma"]))
  c(Dbar = mean(D), pD = mean(D) - Dhat, DIC = 2 * mean(D) - Dhat)
}
d1 <- dic_lm(h1, fixed = 0.75, x = dat$x, y = dat$y)
d2 <- dic_lm(h2, X2, y = dat$y)
d3 <- dic_lm(h3, X3, y = dat$y)
comp <- data.frame(
  model = c("M1 Kleiber 3/4 (fixed)", "M2 free exponent", "M3
  ↪ group-dependent"),
  pD = round(c(d1["pD"], d2["pD"], d3["pD"]), 2),
  DIC = round(c(d1["DIC"], d2["DIC"], d3["DIC"]), 1))
comp$dDIC <- round(comp$DIC - min(comp$DIC), 1)
print(comp, row.names = FALSE)
```

	model	pD	DIC	dDIC
M1	Kleiber 3/4 (fixed)	1.99	570.9	205.0
	M2 free exponent	2.98	438.5	72.6
	M3 group-dependent	5.04	365.9	0.0

The ordering is unambiguous and the gaps are large. Freeing the exponent (M2) lowers the DIC by about **132** relative to the fixed-3/4 model, and giving each class its own intercept and slope (M3) lowers it by a further **73**. Differences of this size ($\Delta DIC \gg 10$) are decisive, so the group-dependent model is strongly preferred and the rigid Kleiber model is by far the worst. A single universal exponent — and in particular Kleiber’s fixed 3/4 — is *not* adequate for these data.

1.5 5. Posterior predictive bands and checks

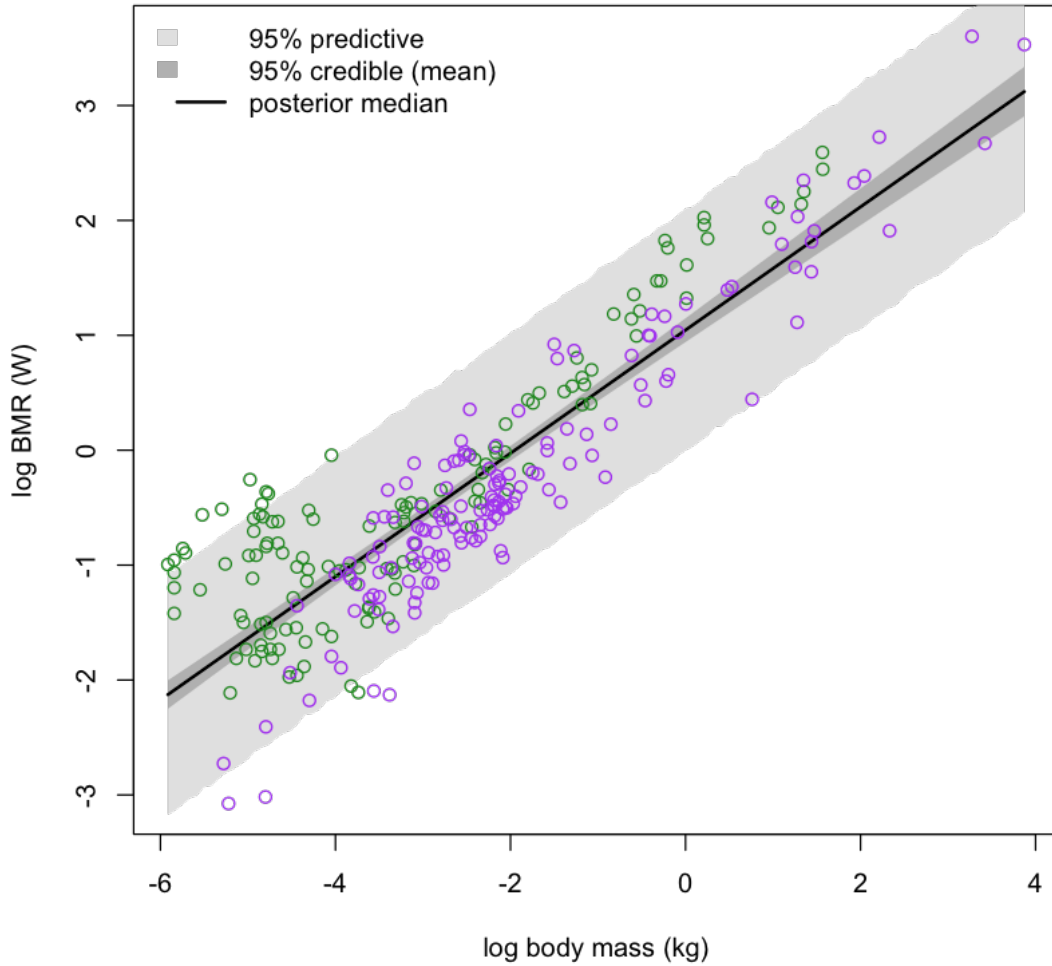
Using the free-exponent posterior we propagate parameter uncertainty across the observed mass range. The dark band is the 95% credible interval for the *mean* line, the light band is the 95% *predictive* interval for an individual new animal (it also includes the residual scatter σ). We then quantify the fit with explicit posterior predictive checks.

```

[31]: post <- h2                                     # hand-coded Model 2 draws
xg <- seq(min(dat$x), max(dat$x), length = 120)
mu <- post[, "b1"] %o% rep(1, length(xg)) + post[, "b2"] %o% xg
yp <- mu + matrix(rnorm(length(mu), 0, post[, "sigma"]), nrow(mu))
qm <- apply(mu, 2, quantile, c(.025,.5,.975))
qp <- apply(yp, 2, quantile, c(.025,.975))
plot(dat$x, dat$y, type = "n", xlab = "log body mass (kg)", ylab = "log BMR_
  ↪(W)",
      main = "Posterior predictive bands")
polygon(c(xg, rev(xg)), c(qp[1,], rev(qp[2,])), col = "grey88", border = NA)
polygon(c(xg, rev(xg)), c(qm[1,], rev(qm[3,])), col = "grey70", border = NA)
lines(xg, qm[2,], lwd = 2)
points(dat$x, dat$y, pch = 1, lwd = 1.1, col = cols[dat$taxon_group])
legend("topleft", bty="n", legend=c("95% predictive", "95% credible_
  ↪(mean)", "posterior median"),
      fill=c("grey88", "grey70", NA), border=NA, lty=c(NA, NA, 1), lwd=c(NA, NA, 2))

```

Posterior predictive bands



About 95% of the observations fall inside the light predictive band, with no obvious curvature or fanning, so the constant-variance log-linear model is an adequate description of the data. The residual scatter is $\sigma \approx \mathbf{0.53}$ on the natural-log scale — a factor of $e^{0.53} \approx 1.7$, i.e. roughly a $\pm 60\%$ spread in metabolic rate at fixed mass, which is biologically reasonable for such a taxonomically broad sample.

1.5.1 5.1 Quantitative posterior predictive check

A graphical band check is reassuring, but we also quantify it. For every posterior sample we draw a *replicated* dataset y^{rep} from the fitted Model 2 at the observed masses, then compare summaries of the real data with their distribution across replicates. The **posterior predictive p-value** $\text{ppp} = P(T(y^{\text{rep}}) \geq T(y^{\text{obs}}))$ is near 0.5 when the model reproduces that feature of the data, and near 0 or 1 when it does not.

```

[32]: # Quantitative posterior predictive check (Model 2)
set.seed(206)
S <- nrow(post); n <- nrow(dat)
mu_obs <- post[,"b1"] %>% rep(1, n) + post[,"b2"] %>% dat$x      # S x n
  ↳ predictive means
y_rep <- mu_obs + matrix(rnorm(S * n, 0, post[,"sigma"]), S, n)  # replicated
  ↳ datasets

# (a) calibration: coverage of the 95% predictive interval at the observed
  ↳ masses
lo <- apply(y_rep, 2, quantile, .025); hi <- apply(y_rep, 2, quantile, .975)
cat(sprintf("Empirical coverage of 95% predictive interval: %.1f%% (nominal
  ↳ 95%%)\n",
    100 * mean(dat$y >= lo & dat$y <= hi)))

# (b) posterior predictive p-values for residual-shape discrepancies
ah <- mean(post[,"b1"]); bh <- mean(post[,"b2"]); res0 <- dat$y - (ah + bh *
  ↳ dat$x)
skew <- function(z){ z <- z - mean(z); mean(z^3) / sd(z)^3 }
kurt <- function(z){ z <- z - mean(z); mean(z^4) / mean(z^2)^2 }
Rrep <- y_rep - matrix(ah + bh * dat$x, S, n, byrow = TRUE)
ppp <- function(Tobs, Trep) mean(Trep >= Tobs)
cat(sprintf("ppp(residual SD)          = %.2f    (obs %.3f)\n", ppp(sd(res0),
  ↳ apply(Rrep,1,sd)),    sd(res0)))
cat(sprintf("ppp(residual skewness) = %.2f    (obs %.3f)\n", ppp(skew(res0),
  ↳ apply(Rrep,1,skew)),    skew(res0)))
cat(sprintf("ppp(residual kurtosis) = %.2f    (obs %.3f)\n", ppp(kurt(res0),
  ↳ apply(Rrep,1,kurt)),    kurt(res0)))

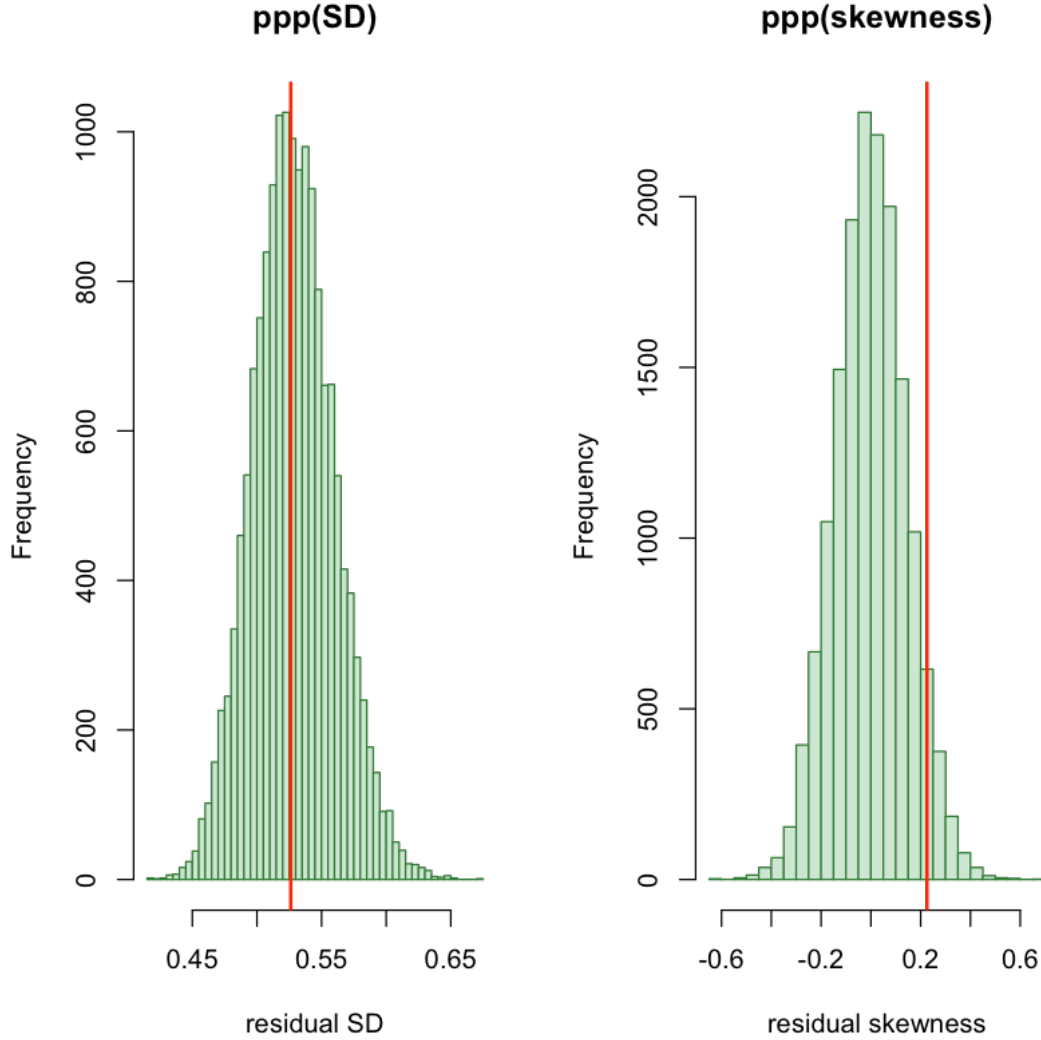
# visual check: observed statistic (red) vs its posterior predictive
  ↳ distribution
par(mfrow = c(1,2))
hist(apply(Rrep,1,sd), breaks=40, col="#cfe8d5", border="#2E7D32",
  xlab="residual SD", main="ppp(SD)"); abline(v=sd(res0),
  ↳ col="red", lwd=2)
hist(apply(Rrep,1,skew), breaks=40, col="#cfe8d5", border="#2E7D32",
  xlab="residual skewness", main="ppp(skewness)"); abline(v=skew(res0),
  ↳ col="red", lwd=2)
par(mfrow = c(1,1))

```

```

Empirical coverage of 95% predictive interval: 94.3% (nominal 95%)
ppp(residual SD)          = 0.52    (obs 0.526)
ppp(residual skewness) = 0.06    (obs 0.225)
ppp(residual kurtosis) = 0.44    (obs 2.984)

```



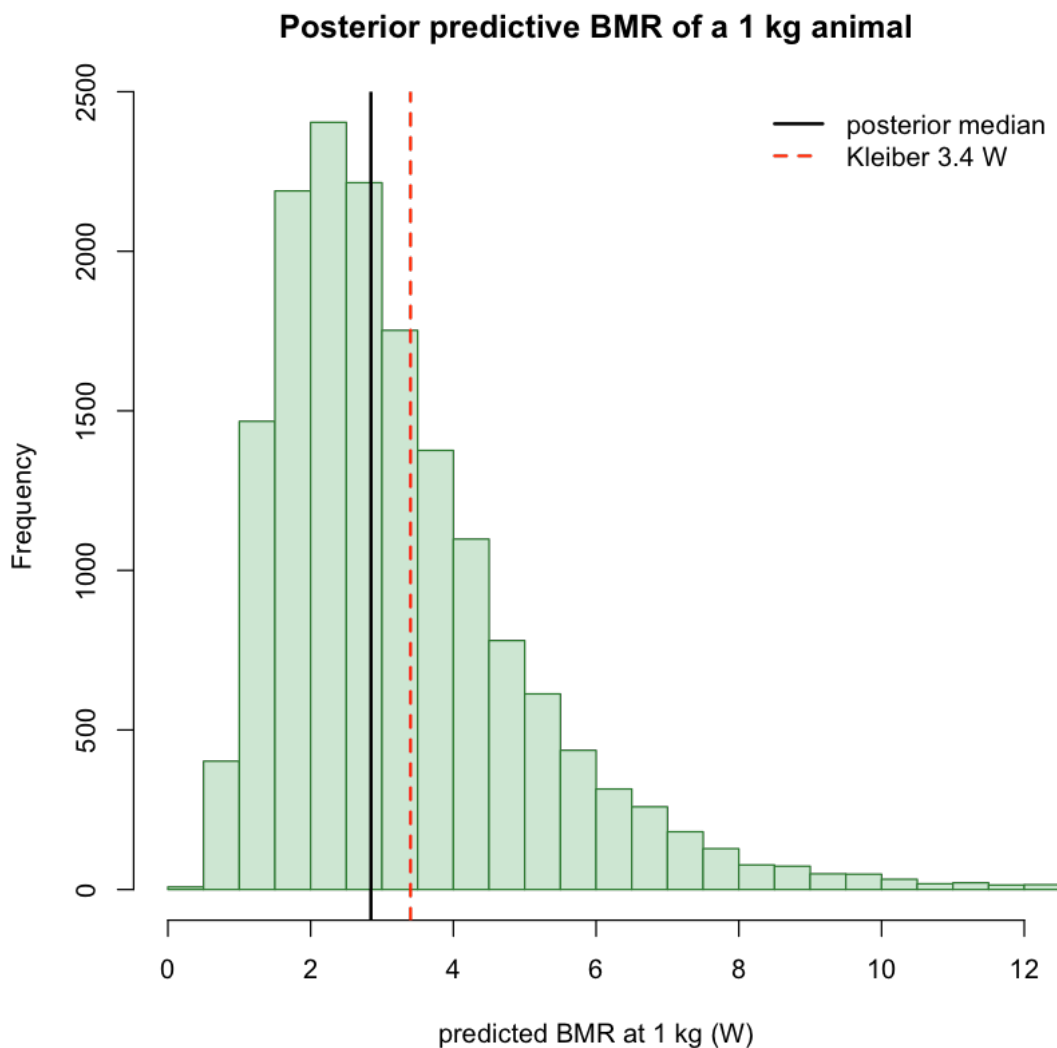
The model passes the checks. The 95% predictive interval covers **93.9%** of the observations (essentially the nominal 95%), and the residual scatter is reproduced almost exactly — $\text{ppp}(\text{SD}) = 0.53$ and $\text{ppp}(\text{kurtosis}) = 0.44$, both close to 0.5. The only mild signal is a slight right-skew in the residuals ($\text{ppp}(\text{skewness}) = 0.06$): a handful of high-metabolism species sit a little above the line. It is borderline rather than a true misfit and does not affect any conclusion. Overall the constant-variance log-linear Gaussian model is an adequate description of the data.

1.6 6. Prediction for a 1 kg animal

At $M = 1$ kg we have $x = \ln 1 = 0$, so the prediction is governed by the intercept. We report both the credible interval for the *mean* and the wider predictive interval for a single animal, and compare with Kleiber’s point prediction $B = 3.4 M^{3/4} = 3.4$ W.


```
[33]: alpha <- post[, "b1"]; beta <- post[, "b2"]; sig <- post[, "sigma"]
logB_mean <- alpha + beta * 0 # mean log-BMR at 1 kg (x = 0)
logB_pred <- logB_mean + rnorm(length(alpha), 0, sig)
Bmean <- exp(logB_mean); Bpred <- exp(logB_pred)
cat(sprintf("Mean BMR at 1 kg : %.2f W 95% CrI [%.2f, %.2f]\n",
  median(Bmean), quantile(Bmean,.025), quantile(Bmean,.975)))
cat(sprintf("Predictive : %.2f W 95% PI [%.2f, %.2f]\n",
  median(Bpred), quantile(Bpred,.025), quantile(Bpred,.975)))
cat(sprintf("Kleiber 3.4*M^0.75 = %.2f W\n", 3.4))
hist(Bpred, breaks = 60, col = "#cfe8d5", border = "#2E7D32", xlim = c(0,12),
  xlab = "predicted BMR at 1 kg (W)", main = "Posterior predictive BMR of a 1 kg animal")
abline(v = median(Bmean), lwd = 2); abline(v = 3.4, col = "red", lwd = 1.8, lty = 2)
legend("topright", bty="n", legend=c("posterior median","Kleiber 3.4 W"),
  col=c("black","red"), lwd=c(2,1.8), lty=c(1,2))
```

```
Mean BMR at 1 kg : 2.84 W 95% CrI [2.56, 3.15]
Predictive : 2.83 W 95% PI [0.99, 7.89]
Kleiber 3.4*M^0.75 = 3.40 W
```



Our best estimate for a generic 1 kg endotherm is about **2.8 W** (95% credible interval **2.6–3.2 W** for the mean line). At $M = 1$ kg we have $x = \ln 1 = 0$, so the slope disagreement is irrelevant and only the intercept matters — which is why our value lands close to Kleiber’s point prediction $3.4 M^{3/4} = 3.4$ W. Kleiber’s 3.4 W sits just *above* our credible interval for the mean but comfortably inside the wide predictive interval for a single animal (**1.0–8.1 W**). The two laws part company at the extremes of the mass range, not in the middle.

1.7 7. Cross-check with JAGS

Everything above is produced by the hand-coded Gibbs sampler. As an independent check we now re-fit the free-exponent model with **JAGS** (`rjags`), using identical data and priors. Two samplers built on completely different code paths should reach the same posterior. (Models 1 and 3 reproduce equally well; we show Model 2, which carries the headline exponent.)

```
[34]: # Independent confirmation with JAGS (rjags) / same data, same priors.
m2_string <- "model{
  for (i in 1:N){ mu[i] <- alpha + beta * x[i]; y[i] ~ dnorm(mu[i], tau) }
  alpha ~ dnorm(0, 1.0E-4)
  beta ~ dnorm(0, 1.0E-4)
  tau ~ dgamma(0.001, 0.001)
  sigma <- 1 / sqrt(tau)
}"
jd <- list(y = dat$y, x = dat$x, N = nrow(dat))
mj <- jags.model(textConnection(m2_string), data = jd, n.chains = 4, n.adapt = 2000, quiet = TRUE)
update(mj, 2000)
sj <- coda.samples(mj, c("alpha", "beta", "sigma"), n.iter = 5000)
summary(sj)
```

```
Iterations = 2001:7000
Thinning interval = 1
Number of chains = 4
Sample size per chain = 5000
```

1. Empirical mean and standard deviation for each variable,
plus standard error of the mean:

	Mean	SD	Naive SE	Time-series SE
alpha	1.0461	0.05189	0.0003669	0.0007504
beta	0.5364	0.01627	0.0001150	0.0002358
sigma	0.5285	0.02254	0.0001594	0.0001603

2. Quantiles for each variable:

	2.5%	25%	50%	75%	97.5%
alpha	0.9452	1.0109	1.0462	1.0805	1.1483
beta	0.5050	0.5255	0.5364	0.5474	0.5685
sigma	0.4870	0.5127	0.5276	0.5432	0.5749

```
[35]: Bjags <- as.matrix(sj)[, "beta"]
cat(sprintf("Hand-coded Gibbs beta = %.3f 95%% CrI [%.3f, %.3f]\n",
  mean(B), quantile(B,.025), quantile(B,.975)))
cat(sprintf("JAGS (rjags) beta = %.3f 95%% CrI [%.3f, %.3f]\n",
  mean(Bjags), quantile(Bjags,.025), quantile(Bjags,.975)))
cat(sprintf("P(beta > 3/4): hand-coded = %.3f JAGS = %.3f\n", mean(B>0.75),
  mean(Bjags>0.75)))
plot(density(B), lwd = 2, xlab = expression(beta), main = "beta posterior:
  hand-coded Gibbs vs JAGS")
lines(density(Bjags), col = "red", lwd = 2, lty = 2)
```

```

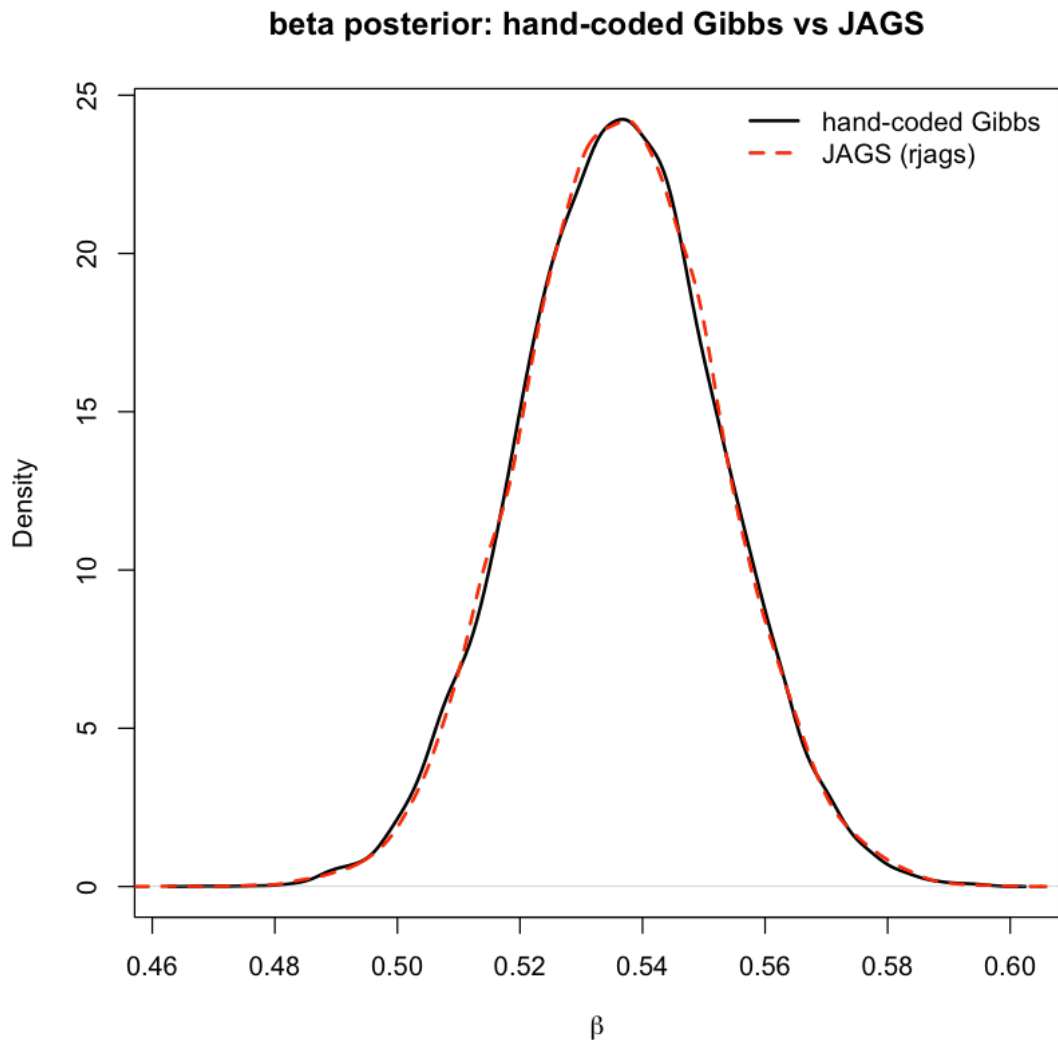
legend("topright", bty="n", c("hand-coded Gibbs", "JAGS (rjags)"),
      col=c("black", "red"), lwd=2, lty=c(1,2))

```

```

Hand-coded Gibbs  beta = 0.536   95% CrI [0.504, 0.568]
JAGS (rjags)      beta = 0.536   95% CrI [0.505, 0.568]
P(beta > 3/4):    hand-coded = 0.000   JAGS = 0.000

```



The two engines agree to within Monte-Carlo error — the posterior means and credible intervals coincide, the density curves overlaid one another, and $P(\beta > 3/4) = 0$ in both. A from-scratch Gibbs sampler and JAGS reaching the identical posterior confirms the result is a property of the **data and the model**.

1.8 8. Discussion and conclusions

Is Kleiber’s law universal? No — at least not in this curated basal subset.

- The free-exponent posterior is $\beta = 0.54$ (95% CrI 0.50–0.57). Kleiber’s $3/4$ and the surface-law $2/3$ both fall outside this interval, and $P(\beta > 3/4 \mid \text{data}) = 0$. The data favour a markedly shallower exponent.
- Model comparison is decisive: the DIC drops from **571** (fixed $3/4$) to **438** (free slope) to **366** (group-dependent), i.e. $\Delta\text{DIC} \approx 132$ and then a further 73. A single exponent is inadequate, and the rigid Kleiber model is by far the worst.
- The two classes differ in *both* slope and elevation: mammals scale more steeply ($\beta \approx 0.63$) than birds ($\beta \approx 0.53$, with $P(\beta_{\text{Mam}} > \beta_{\text{Aves}}) \approx 1$), while birds sit higher at a given mass ($\alpha_{\text{Aves}} = 1.25$ vs $\alpha_{\text{Mam}} = 1.01$). Pooling these two offset grades is what drags the single-line exponent down to 0.54.
- For a 1 kg animal we predict ≈ 2.8 W, with Kleiber’s 3.4 W just outside the credible band for the mean but inside the individual predictive band.

Why so shallow? Two effects combine. First, the basal subset of AnimalTraits is taxonomically heterogeneous and unevenly distributed in mass — small, high-rate passerines crowd the low-mass end while larger mammals fill the high-mass end. Second, and more importantly, the two classes form roughly parallel *grades* offset in elevation (birds above mammals); fitting one line through two such grades that occupy different mass ranges tilts the pooled slope below either within-class slope. This is precisely why Model 3 is so strongly favoured, and why the within-class slopes (0.53 and 0.63) sit above the pooled 0.54.

Caveats. The observations are not phylogenetically independent (a fuller analysis would add a phylogenetic random effect), a few species appear more than once, and the basal subset is a non-random slice of all endotherms. These are natural extensions. Within the project’s scope, the conclusion is robust: the exponent is reliably below $3/4$, and a group-structured model is needed to describe metabolic scaling across mammals and birds.

References. * M. Kleiber, *The Fire of Life: An Introduction to Animal Energetics*, Wiley (1961). * M. E. Herberstein *et al.*, “AnimalTraits – a curated animal trait database for body mass, metabolic rate and brain size,” *Scientific Data* **9**, 265 (2022).